



THEME: ACCESSIBILITY, ECOMMERCE & CYBER SECURITY

AaaS — Accessibility as a Service

Accessibility infrastructure for public institutions: Odisha first, India next

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One button on any government website that reads, translates, simplifies, reads scanned documents and listens: in Odia, by voice, with no cloud.

1 The problem

Pensions, scholarships, exam results, grievances: Odisha's public services have moved online. Most of the citizens who need them most have not. The websites were built for people who read English easily, can see small text, and can type.

4.2 crore

people in Odisha (Census 2011)

3.75 crore

speak Odia as their mother tongue (Census 2011)

27%

of Odisha's people aged 7+ cannot read (literacy 72.9%, Census 2011)

2.68 crore

Indians live with a disability (Census 2011)

English-first pages

Notices and forms are written in English and in heavy official language.

Scanned notices

Important circulars are photos of paper. A screen reader or translator sees nothing.

Typing-only forms

Elderly people, people with shaky hands and people who cannot read are locked out of applying.

No Odia voice

Common tools cannot speak Odia well. Most cannot even say Odia numbers.

Arun is 68, a farmer near Jajpur. He has arthritis in both hands and reads only a little Odia. His pension rate just went up, but the order is a scanned English circular and the application form wants typed answers. His government is online. Arun is not. His grandson has to do it for him.

2 Our solution

AaaS puts one round ଅ button on any website. Tapping it opens six tools, labelled in Odia first, that let a citizen **listen to, understand and fill in** the page, without the website being rebuilt.

For a government website

The website owner adds **one line of code**. Every visitor gets the button. One shared server (for example at OCAC) can serve every department.

For any website, today

Our **Chrome extension** adds the same button to sites that haven't adopted AaaS yet. We tested it on the real Jajpur district and SSEPD department websites.

3 Six tools behind one button

TOOL	WHAT HAPPENS	WHO IT HELPS
ପଢ଼ି ଶୁଣାଅ Read aloud	The page is read aloud in a natural Odia voice. Each part is highlighted as it is spoken, so the reader can follow. Skip, replay and pause buttons. Numbers are spoken as Odia words (“୩୦୦ ଟଙ୍କା” → “three hundred rupees” in Odia).	blind low vision cannot read elderly
ଅନୁବାଦ Translate	The whole page turns into Odia in place, keeping its layout. Dates, amounts and reference numbers stay exactly correct. One tap (↵) brings back the original.	Odia speakers rural citizens
ସହଜ ପଢ଼ା Easy Read	Heavy official language becomes simple words: “shall furnish the requisite documents” → “must give the papers”. Works in English or Odia.	low literacy learning difficulty
ଦଲିଲ ପଢ଼ Read document	Finds scanned notices and PDFs on the page and reads the text off them (OCR). It turns the text into Odia and shows the original page and the Odia text side by side , reading aloud with a highlight that follows the voice.	blind cannot read everyone
କହି ଲେଖ Speak to fill	Goes through a form field by field, asks each question aloud in Odia and listens to the answer. Names are written in English letters, not translated. Spoken numbers become digits. It never presses Submit : the citizen decides.	motor disability elderly cannot type
କହି ଚଳାଅ Voice command	Say where you want to go (“ବିଜ୍ଞପ୍ତି” for notifications) and it opens that link, even links hidden inside menus. If it isn’t sure, it shows the top choices as buttons instead of guessing.	blind motor disability

Also included: ଆରାମ ଅକ୍ଷର Comfortable letters (OpenDyslexic font, extra spacing, warm background) for dyslexic readers, a reading ruler, bigger text and speech for any text you select or point at.

4 Two journeys it makes possible

Arun applies for his pension, alone

- Taps ଦଲିଲ ପଢ଼: hears the scanned pension order read out in Odia, with the date and amount correct.
- Taps କହି ଲେଖ: speaks his name “ପୂର୍ଣ୍ଣଚନ୍ଦ୍ର ମହାନ୍ତି”, and the form shows *Purnnachandra Mahanti* (not “full moon”, which is what word-for-word translation gives).
- Says his phone number as “ନଅ ଆଠ ସାତ... ଡବଲ ପାଞ୍ଚ” and gets 98755...
- Checks the ✓ marks and presses Submit himself.

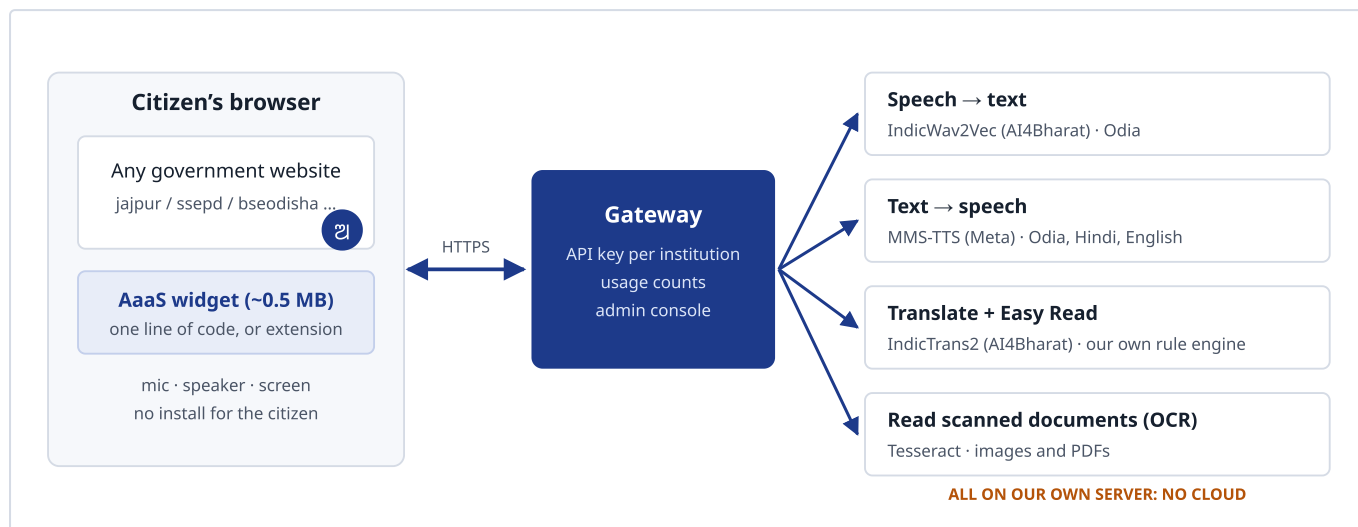
A blind student checks an exam circular

- Opens the BSE Odisha site. The circular is a scanned image.
- Voice command: “ବିଜ୍ଞପ୍ତି”. The notices page opens.
- ଦଲିଲ ପଢ଼ reads the circular in Odia, with the exam dates spoken correctly.
- No sighted helper and no paid software needed.

Designed for the people who use it

Big icon tiles with Odia labels first · works with keyboard only (Space pauses, Esc stops) · our own panel follows accessibility rules (screen-reader labels, strong contrast, size set so the host page can’t shrink it) · every action can be undone · the voice never submits anything for you.

5 Architecture



The widget is plain JavaScript with no outside libraries, kept separate from the host page so it cannot break the page and the page cannot break it. The four AI services are open-source models running on a normal laptop CPU (no graphics card needed). Every service has a backup mode, so if one fails the button stays alive and says so.

6 Hard problems we solved

1 The Odia voice model is silent on numbers.

We convert “୩୦୦ ଟଙ୍କା” or “12.05.2026” into Odia words before speaking, so amounts and dates are actually heard.

2 Translation changes names and dates.

Word-for-word translation turns the name Purnnachandra into “full moon”, so for names we write the Odia sounds in English letters instead of translating the meaning. The translation model also changed dates on long notices (“2026” came back as “2021”), so we hide every number from the model and put the originals back afterwards.

3 People say English menu names in Odia.

The speech model writes “Contact” as “କଣ୍ଟାକ୍ଟ”. We compare how words sound, so an Odia speaker can open English-named links by voice, including ones hidden in menus.

4 Spoken numbers.

“ନଅ ଆଠ ସାତ ଡବଲ ପାଞ୍ଚ” or “ଷାଠିଏ ପାଞ୍ଚ” become 98755 and 65 in the form field.

5 Scanned documents.

Finds images and PDFs on the page, lets the user pick one by its thumbnail, reads the text, then simplifies, translates and speaks it, page by page.

6 Real government websites.

They block microphones, hide links and mark pages “do not translate”. We handle each case and tell the user clearly when a website itself blocks something.

Privacy and safety by design

- Voice, documents and form answers never leave the government’s own server: no cloud AI, and it fits India’s DPDP Act.
- Easy Read uses rules, not a chatbot, so it can never invent a wrong deadline.
- It never submits a form on its own.

Cheap to run, free to adopt

- Runs on one ordinary laptop or server CPU, and works fully offline.
- Open source (Apache-2.0): no licence fee, and Odisha owns it.
- Repeated pages are served from a cache, so they are instant.

7 Why AaaS is different

EXISTING OPTION	WHAT IT GIVES	WHAT AAAS ADDS
Bhashini (national language AI)	Language engines as APIs: translation, speech.	The finished citizen-facing product on top: guided voice form-filling, name handling, scanned-notice reading, voice navigation. Bhashini could be plugged in as an engine.
Browser read-aloud / translate	Generic reading and translation, usually through the cloud.	Odia numbers spoken correctly, OCR of scanned circulars side by side, form filling by voice, fully offline.
Accessibility “overlays”	Plug-ins that claim to make a site compliant.	We claim nothing about the host site. We give the citizen a working way through it, and we are open source and auditable.
Rebuilding every portal	Years of work and budget per department.	One line of code per website, one shared server for the whole state.

8 Rollout plan

STEP 1 Pilot One district website (Jajpur) and one board site (BSE Odisha). One line of code each.	STEP 2 OCAC server One shared install at the state data centre, with an API key per department.	STEP 3 All of Odisha Every department portal, in line with 5T and Mo Sarkar. The extension covers the rest.	STEP 4 India The engines already cover 22 Indian languages: add a language, not a product.
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9 Where it stands today

Built and working (live demo)

- ✓ All six tools and the reading-comfort options
- ✓ Chrome extension, tested on the real jajpur.odisha.gov.in and ssepd.odisha.gov.in websites
- ✓ Demo copies of the SSEPD and BSE Odisha websites with real scanned notices
- ✓ Four AI services and a gateway, running fully offline on one laptop
- ✓ 240+ automated tests passing
- ✓ Admin console showing usage for each institution

Next steps

- More Indian languages (Hindi and English voices already work)
- Better OCR for poor-quality or handwritten scans
- Tuning speech recognition for noisy places and village accents
- Accessibility-check reports for website owners
- Sign-language avatar (early concept only)

Honest limits today: printed scans work well and handwriting does not; speech needs a reasonably quiet room, which is why the tool asks again or offers choices when unsure.

10 Expected impact

Independence

Elderly, disabled and non-reading citizens use online services without a middleman or a fee

Reach

Every existing Odisha portal becomes Odia-speaking, with no redesign

Trust

Scanned notices become understandable; data stays in Odisha

**“Your portals stay exactly as they are.
Your citizens stop needing help to use them.”**

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Guide: Somali Priyadarshini Mohanty
github.com/flashmindbox/AaaS